

A Machine-Checked Finite Reduction for Erdős Problem #287 and a Source-Explicit Large- M Criterion

Clean Public Review + Self-Contained Gate Theory Manual

Version 16.3 – Prepublication Candidate R4 (not yet published)

Research draft with Lean/Aristotle-assisted formal verification

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Status. Erdős Problem #287 remains open. The unconditional public Lean result excludes every exact counterexample with maximum denominator at most 4×10^9 , and the large- M conclusion is compiled from the explicit `WindowPairSupply` interface. A later V15 Aristotle replay, supplied with this audit, reports a successful 8118-job kernel build of the source-minimal identity $\Lambda = \mu * \log$, its affine source, the determinant-one line, the balanced-seven coefficient -20 , and the squarefree labelled polarization. Public repository `main` already contains individual V15 source modules, but its aggregate `Main.lean/status` replay is not yet synchronized; this note therefore distinguishes *Lean-proved in the latest supplied replay* from *public-main synchronized*. The latest supplied V16 factorial Aristotle replay reports the factorial Euler seven-box polarization, its repeated-prime fibre identity, the local Euler/generalized-von-Mangoldt algebra, expected-term linearity and the Pascadi parameter ledger as kernel-checked in an 8123-job build. Public-main synchronization remains a separate provenance question. The first exact balanced-seven analytic open is `AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45`; the physical comparison match, remaining packets, large- M supply, and Problem #287 remain open. This R4 candidate has not been pushed to the stable public PDF.

Abstract

Erdős Problem #287 asks whether every representation

$$1 = \frac{1}{n_1} + \cdots + \frac{1}{n_k}, \quad 1 < n_1 < \cdots < n_k,$$

contains a consecutive gap of size at least 3. The present project has one unconditional machine-checked result and one explicit conditional large- M interface. Lean excludes all exact counterexamples with maximum denominator at most 4×10^9 , while an effective eventual `WindowPairSupply` theorem would close the remaining range.

For the analytic programme we use the source-minimal identity

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r, \quad s = \pm 1,$$

together with its exact small- q /small- r /hard partition and determinant-one affine line. A supplied V15 Aristotle replay reports these identities, the balanced-seven coefficient -20 , and the earlier squarefree polarization as kernel-checked. The current balanced-seven representation is sharper: for

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p)$$

we introduce the factorial Euler multiplicative function

$$F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}.$$

Its degree- $(1, \dots, 1)$ torus coefficient recovers the ordered seven labelled prime-box source *including repeated primes exactly*; hence the previous $X^{6/7+o(1)}$ collision remainder is no longer part of the controlling formula. Polarization also commutes with every linear expected-term operator. The remaining balanced-seven inputs are therefore (i) one signed factorial endpoint distribution estimate at $(Q, y) = (X^{3/5}, X^{1/7})$, and (ii) identification of the factorial expected term with the physical comparison/main term.

The endpoint estimate required here is strictly narrower than a Bombieri–Vinogradov theorem for all class- $\mathcal{C}$ functions, an absolute modulus average, a theorem uniform in $\mathbf{z}$, or a general smooth-number theorem. A direct audit of Pascadi’s current arXiv v2 proof parameters is nonclosing for this particular route: the proof architecture uses $Q \leq x^{5/8-100\eta}$ together with $y \leq x^\eta$; setting $Q = x^{3/5}$ forces $\eta \leq 1/4000$, whereas $y = x^{1/7}$ requires $\eta \geq 1/7$. This is a limitation of that written proof route, not a theorem excluding future extensions. The paper also states the Gate 0–2 theory itself in theorem form, so that each source, target, pin and non-implication can be audited without the historical dossier. No analytic endpoint estimate, comparison identification, eventual large- M supply, or proof of Erdős #287 is claimed.

Status at a glance

Item	Status	Meaning
Exact public counterexample predicate	PUBLIC LEAN-CHECKED	Exact set formulation and ordered public statement are connected.
Finite range $M \leq 4 \times 10^9$	PUBLIC LEAN-CHECKED	A certificate chain excludes every exact counterexample in this range.
Window-pair closure compiler	PUBLIC LEAN-CHECKED / CONDITIONAL	Eventual effective <code>WindowPairSupply</code> plus the finite theorem implies #287.
$\Lambda = \mu * \log$, affine source, determinant-one line	LEAN-PROVED IN LATEST V15 REPLAY	User-supplied V15 report records an 8118-job kernel build; public-main synchronization is still pending.
Balanced-seven –20 and squarefree polarization	LEAN-PROVED IN LATEST V15 REPLAY	Finite/algebraic only; no analytic cancellation follows.
Factorial Euler seven-box polarization	LEAN-PROVED IN LATEST V16 FACTORIAL REPLAY	Exact ordered-source encoding including repeated primes; analytic endpoint remains open.
Expected-term/polarization linearity	RESEARCH/ALGEBRAIC PASS	Coefficient extraction commutes with every linear expected-term operator.
Gate 0	MIXED / NOT CLOSED	$b.1$ and growth are controlled; uniform $b.2$, the literal Type-I source theorem and comparison/source pins remain open or external.
Gate 1A	CONDITIONAL POSSIBLE PROVIDER	Canonical common-weight mean-square theorem under explicit source pins; current source/fibre instantiation remains open and its necessity for #287 is census-dependent.
Gate 1B	OPEN COMPLEMENTARY PROVIDER	Centered/Möbius source algebra is banked; a source-to-convolution-BV/provider theorem remains open.
Ford generated-packet census	OPEN SOURCE/REASSEMBLY	Universal Type II is stronger than the proof-specific family actually generated at Ford–Maynard (7.23).
Gate 2	CONDITIONAL ENDGAME / NOT ACTIVATED	Consumes Gate-0 data, a complete Type-II package, the N_2 replacement and compatible ε to yield positive prime mass.
Direct Pascadi v2 proof route at (3/5, 1/7)	AUDITED NONCLOSING	Written parameters force $\eta \leq 1/4000$ but the source requires $\eta \geq 1/7$.
Factorial signed endpoint estimate	OPEN ANALYTIC	FACTORIAL-OMEGA7-SIGNED-ENDPOINT45 (full programme label given in Section 17.4).
Physical comparison identification	OPEN SOURCE	Principal local density plus the literal exceptional-character convention must match the physical comparison term.
Remaining packets / FCL / eventual supply	OPEN	Balanced-seven closure alone does not finish the large- M programme.
Erdős Problem #287	OPEN	No unconditional solution claim.

Status vocabulary. PUBLIC LEAN-CHECKED means present on public repository `main`. LEAN-PROVED IN LATEST V15 REPLAY means the supplied Aristotle V15 build report records a successful kernel replay, while public-main synchronization remains a separate provenance step. LEAN-PROVED IN LATEST V16 FACTORIAL REPLAY means the exact mathematical derivation has passed the current hostile audit but no completed Lean replay is claimed. OPEN ANALYTIC and OPEN SOURCE denote deliberately unproved inputs. AUDITED NONCLOSING refers only to the literal cited theorem/proof dictionary, not to a statement that no future extension can work.

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1 The problem and the scope of this note

The public problem asks whether, for distinct integers

$$1 < n_1 < \cdots < n_k, \quad \sum_{i=1}^k \frac{1}{n_i} = 1,$$

one necessarily has

$$\max_{1 \leq i < k} (n_{i+1} - n_i) \geq 3.$$

It is listed as open in the Erdős Problems archive and originates in the sources cited there [1, 2].

For a finite set $A \subset \mathbb{N}$, write

$$N = \min A, \quad M = \max A.$$

A counterexample is a finite set satisfying

$$A \subset \{2, 3, \dots\}, \quad \sum_{a \in A} \frac{1}{a} = 1,$$

and the local gap condition that every nonmaximal $a \in A$ is followed by $a + 1$ or $a + 2$ in A . The Lean development formalizes exactly this predicate and proves the ordered form from it.

This note is deliberately not a chronological research diary. It records only the current controlling finite theorem, the exact conditional closure interface, the source-minimal algebra, the factorial Euler balanced-seven reduction, and the current open analytic obligations. Earlier squarefree/collision routers, Vaughan decompositions, failed provider dictionaries, hostile audits, and superseded residuals belong to the separate technical verification ledger and immutable V15 archive.

2 The machine-checked finite theorem

Theorem 2.1 (Finite exclusion). *There is no exact counterexample to Erdős Problem #287 with*

$$3 \leq M \leq 4,000,000,000.$$

Equivalently, every unit-fraction representation in the problem with maximum denominator at most 4×10^9 has a consecutive gap at least 3.

The theorem is represented in the repository by `no_Erdos287Counterexample_of_max_le_4e9`. Its proof is not an exhaustive enumeration of finite subsets. It is a certificate compiler built from the three ideas below.

2.1 Holes and the adjacent-hole contradiction

For a hypothetical counterexample, call an integer $x \in [N, M]$ a *hole* when $x \notin A$. The $\text{gap} \leq 2$ condition implies that two consecutive integers in $[N, M]$ cannot both be holes. This is the formal theorem `Gap2CE.holes_isolated`.

A second elementary fact places the top half of the interval above the minimum:

$$N \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Thus, whenever $M \leq 2x$ and $x + 1 \leq M$, the consecutive pair $x, x + 1$ lies inside the closed window where the adjacent-hole contradiction applies.

2.2 Prime-power windows

For $j \geq 0$, let $C(j)$ denote the largest reduced numerator arising from a nonempty partial reciprocal sum

$$\sum_{s \in S} \frac{1}{s}, \quad \emptyset \neq S \subseteq \{1, \dots, j\}.$$

The formal development uses a certified upper table

$$C_{\text{val}}(0), \dots, C_{\text{val}}(9) = 1, 1, 3, 11, 25, 137, 137, 1019, 2143, 7129.$$

The prime-power window theorem says, schematically, that if p is prime, $e \geq 1$, $p^e \mid x$, and

$$C\left(\left\lfloor \frac{M}{p^e} \right\rfloor\right) < p,$$

then x cannot belong to A . The reason is that the top p -adic layer of the reciprocal sum would force a nonzero numerator to vanish modulo p .

Consequently, if both x and $x + 1$ possess such prime-power divisors, then they are two adjacent holes. This gives the central finite compiler.

Proposition 2.2 (Window-pair blocker). *Suppose there are primes p_u, p_v , positive exponents a_u, a_v , and an integer x such that*

$$\begin{aligned} p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor, \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \\ C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \end{aligned}$$

and

$$M \leq 2x, \quad x + 1 \leq M.$$

Then a gap- ≤ 2 counterexample with maximum M cannot exist.

2.3 The interval-certificate chain

One certificate can exclude an entire interval $L \leq M \leq U$. The divisibility data are fixed at a chosen x , monotonicity controls the windows as M varies, and $U \leq 2x$ keeps the forced holes above the minimum. The repository contains a chain of 34 certificates whose intervals meet end-to-end and cover

$$[3, 4 \times 10^9].$$

All primality side conditions are discharged in the kernel; the construction does not use `native_decide`.

2.4 Sign-sensitive Sophie witnesses and the free maximum blocker

The latest supplied finite bank sharpens the final analytic-to-finite landing and is worth stating because it prevents small endpoint mistakes later. A *plus* Sophie witness at M is a prime q with

$$q \geq 5, \quad M < 3q, \quad 2q + 1 \leq M, \quad 2q + 1 \text{ prime},$$

equivalently $M/3 < q \leq (M - 1)/2$ with the integer endpoints understood literally. A *minus* witness is

$$q \geq 5, \quad M < 3q, \quad 2q \leq M, \quad 2q - 1 \text{ prime},$$

so the correct minus endpoint is

$$q \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Either witness contradicts an exact counterexample. The hypothesis $q \geq 5$ is not cosmetic: for a prime it is the same as $q > 3$, which is the finite window threshold needed by the cofactor-two obstruction.

There is also a free maximum-divisor blocker. If $q > 3$ is prime, $q \mid M$ and $M < 3q$, then necessarily $M \in \{q, 2q\}$; in either case the top-layer/cofactor-two exclusion contradicts $M \in A$. Thus a counterexample cannot have prime maximum and cannot have $M = 2q$ with $q > 3$ prime. These are finite repairs, not analytic inputs; they should be used when translating positive affine mass into an actual blocker.

3 The exact large- M closure criterion

Definition 3.1 (Window-pair supply). For $M \in \mathbb{N}$, $\text{WindowPairSupply}(M)$ is the assertion that there exist $x, p_u, a_u, p_v, a_v \in \mathbb{N}$ such that

$$\begin{aligned} p_u, p_v \text{ are prime,} \quad a_u, a_v \geq 1, \\ p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x+1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \\ \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \\ M \leq 2x, \quad x+1 \leq M. \end{aligned}$$

In plain terms, two consecutive integers in the top half of $[1, M]$ each have a prime-power divisor larger than roughly $M/10$, with a small local numerator condition.

Theorem 3.2 (End-to-end conditional compiler). *Assume there is an explicit $M_0 \leq 4 \times 10^9$ such that*

$$\text{WindowPairSupply}(M) \quad \text{for every } M \geq M_0.$$

Then Erdős Problem #287 is true.

This implication is machine-checked as `no_Erdos287Counterexample_of_closure`. The proof splits at 4×10^9 : the finite theorem handles the lower range, while the window-pair blocker handles every larger maximum.

The interface is weaker than a Sophie-Germain prime condition. For example, if a prime q in the interval $M/3 < q \leq (M-1)/2$ has $2q+1$ prime, then one may take $x = 2q$. The analogous minus case uses $x = 2q-1$. The Lean theorem `windowPairSupply_of_sophieWitness` formalizes this implication. The present programme does not claim such Sophie witnesses for every large M .

Part I

Theory and Verification Manual: Gate 0–2

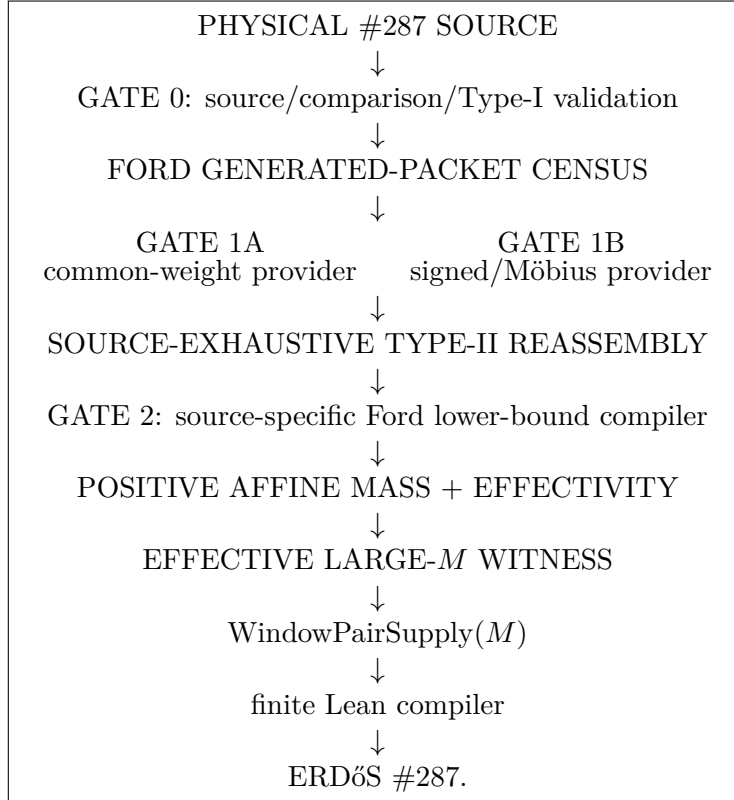
4 What the Gate theory is trying to do

This Part is written for a reader who has never seen the earlier chats, the V15 patch stack, or the repository. The Gate labels are not theorem substitutes: each Gate is a literal mathematical package with an input source, a target theorem, a verification boundary, and specified downstream consequences.

Conceptually, the programme separates four jobs.

- Gate 0.** Construct and validate the physical sequence/comparison class: prime mass, comparison regularity, growth, Type I, and the exact expected-term convention.
- Gate 1.** Supply parity-breaking Type-II estimates for the actual generated packets.
- Gate 1A.** One canonical common-weight moving-source provider.
- Gate 1B.** A complementary signed/Möbius/high-order provider that preserves the arithmetic sign resource through completion/dispersion.
- Gate 2.** Consume a source-exhaustive Type-II package, the Ford arithmetic weight, the exceptional N_2 sector, and one compatible ε_* to run the Ford–Maynard lower-bound endgame.

The intended dependency graph is



(Theory-DAG)

This is a dependency graph, not a proof.

The following non-implications are load-bearing:

Gate 1A closed $\nRightarrow$ Gate 1B closed, Gate 1B closed $\nRightarrow$ Gate 1A closed,
 Gate 1A + Gate 1B $\nRightarrow$ Full Type II without source exhaustiveness,
 generated Type II $\nRightarrow$ Gate 2 without Gate-0/ N_2 /source inputs,
 Balanced7 closed $\nRightarrow$ FCL closed,
 FCL closed $\nRightarrow$ WindowPairSupply(M),
 none of the preceding statements alone proves Erdős #287.

(Theory-firewall)

4.1 Verification vocabulary

Every major object below is assigned one of the following meanings.

Label	Meaning
LEAN-PROVED	Proved by a cited/current Lean kernel replay.
PROVED-FINITE / PROVED-ALGEBRAIC	Exact finite/combinatorial/algebraic mathematics; may or may not also be Lean formalized.
EXTERNALLY-AUDITED	A sourced mathematical theorem or source audit, not formalized here.
CONDITIONAL-COMPILER	The implication is proved, but one or more explicit antecedents remain open.
SOURCE-OPEN	The physical/source identification, normalization or dictionary is missing.
ANALYTIC-OPEN	The source object is sufficiently fixed; a deep estimate remains unproved.
SOURCE-BLOCKED	The literal source object/prefactor needed even to state the final theorem is not yet printed.
CAPACITY-ONLY	Exponent arithmetic passes, but no analytic theorem follows.
FALSE / RETIRED	The literal proposed route is disproved or withdrawn.
SUPERSEDED	Historically valid but no longer the controlling representation.

5 Gate 0: the physical sequence/comparison theorem

5.1 The physical #287 sequence

Fix a nonnegative smooth weight W supported inside the analytic blocker band; the finite blocker itself only needs the corresponding interval to lie in the top half. Define

$$\mathfrak{S}_2 = \prod_{\ell > 2} \frac{\ell(\ell-2)}{(\ell-1)^2}, \quad B(n) = \mathfrak{S}_2 \prod_{\substack{\ell|n \\ \ell > 2}} \frac{\ell-1}{\ell-2},$$

and

$$a_X(n) = W(n/X)(\Lambda(2n-1) + \Lambda(2n+1)),$$

$$b_X(n) = 4W(n/X)B(n), \quad w_X(n) = a_X(n) - b_X(n). \quad (\text{G0-source})$$

The role of b_X is to model the expected local density of the two affine prime forms; w_X is the discrepancy on which Type I/II cancellation is sought.

The elementary comparison algebra includes

$$B(n) = \mathfrak{S}_2 \sum_{\substack{e|n \\ e \text{ odd squarefree}}} \prod_{\ell|e} \frac{1}{\ell-2},$$

and for each fixed d ,

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \frac{d}{\phi(2d)}. \quad (\text{G0-Bmean})$$

Thus one sign has mean $2d/\phi(2d)$ and the two-sign model has mean $4d/\phi(2d)$ at the fixed- d level. Also

$$B(n) \ll \log \log(3n), \quad |w_X(n)| \ll \log X,$$

and, in the present source class,

$$\sum_{n \lesssim X} |w_X(n)| \tau(n) \ll X (\log X)^2. \quad (\text{G0-growth})$$

5.2 Gate-0 target package

Definition 5.1 (Gate-0 package). Gate 0 is *closed for the #287 source at scale X* if the following seven items hold with one common physical normalization.

G0.1 $a_X(n) \geq 0$ and the comparison sequence b_X belongs to the Ford–Maynard ordinary sequence class used in the lower-bound theorem.

G0.2 Prime mass:

$$\sum_p b_X(p) \gg \frac{X}{\log X}.$$

G0.3 Growth/divisor control, including a bound of the form (G0-growth) (or the exact stronger divisor-weight norm required by the chosen Ford invocation).

G0.4 Comparison regularity: the physical b_X satisfies the Ford–Maynard condition (b.2) uniformly over every coagulation region actually used downstream. At the fixed- d level this is compatible with (G0-Bmean); the required uniform publication-level statement must be source-pinned.

G0.5 Literal maximal/weighted Type I:

$$\sum_{m \leq X^{1/2-\varepsilon}} \tau(m)^A \max_I \left| \sum_{\substack{X/2 < mn \leq X \\ n \in I}} w_X(mn) \right| \ll_{A,W,\varepsilon} X (\log X)^{-A}. \quad (\text{G0-I})$$

G0.6 One fixed principal/small-conductor/exceptional-character expected-term convention is used consistently by both the prime source and the comparison source.

G0.7 Even, nonunit, prime-power and boundary strata created by the affine forms $2n \pm 1$ are either included in the source theorem or routed with a proved error.

Current verification. Items G0.1–G0.3 have elementary/source-level proofs in the dossier; the exact finite problem statement and finite blocker are Lean-proved. G0.4 and G0.5 are **SOURCE-OPEN / EXTERNALLY-AUDITED TARGETS** at the exact publication normalization; G0.6 is the same physical comparison convention that later appears in the factorial balanced-seven and Pure5 comparison pins.

What Gate 0 closing buys. It supplies the non-parity hypotheses needed by the Ford lower-bound compiler and removes the possibility that a later Type-II estimate is being applied to the wrong comparison sequence.

What it does not buy. Gate 0 does not prove Type II, parity breaking, Gate 1A, Gate 1B, Gate 2, a large- M witness, or Erdős #287.

6 Gate 1A: canonical common-weight Type-II provider

6.1 Parameters, rows and physical source

At an auxiliary scale X_1 , set

$$\begin{aligned} M &= X_1^{1/3}, & R &= X_1^a, & L &= X_1^b, \\ H &= X_1^{a+2b-2/3}, & K &= \frac{M}{R}, & U &= \frac{L}{H}, & D &= \frac{L^2}{H}. \end{aligned} \quad (\text{G1A-scales})$$

Thus

$$RK = M, \quad DH = L^2, \quad M^2H = RL^2.$$

A row is

$$e = (r, k, m), \quad r \asymp R \text{ prime}, \quad 1 \leq |k| \leq K, \quad m \asymp M, \quad m' = m + kr \asymp M.$$

Choose the canonical affine forms $A_e(t), B_e(t)$ so that

$$\boxed{m'A_e(t) - mB_e(t) = 2k.} \quad (\text{G1A-det})$$

For

$$\rho_\ell(y) = \mathbf{1}_{\ell|y} - \frac{1}{\ell}$$

and bounded prime coefficients b_p, d_q , the *one common physical-weight* source is

$$\boxed{C_e(b, d) = \sum_{t \in \mathbb{Z}} W_D(t) \left(\sum_{p \asymp L} b_p \rho_p(A_e(t)) \right) \left(\sum_{q \asymp L} d_q \rho_q(B_e(t)) \right).} \quad (\text{G1A-source})$$

The word “common” is literal: W_D is one datum. Arbitrary independently chosen weights $W_{D,e}$ are a stronger source class and are not covered by the canonical theorem.

6.2 Gate-1A target theorem

Theorem 6.1 (Canonical common-weight Gate-1A provider – conditional form). *Assume the five source/analytic pins G1A-A through G1A-E below. Then, uniformly for the canonical all- m row family $\mathcal{E}^\sharp$,*

$$\boxed{\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \left(\sum_{p \asymp L} |b_p|^2 \right) \left(\sum_{q \asymp L} |d_q|^2 \right).} \quad (\text{G1A-target})$$

The exact normalized variant used by the formal compiler is equivalent after the banked H^2 normalization bridge.

The theorem is a **CONDITIONAL-COMPILER**: the finite/operator part is substantially banked, but the physical source pins are not all inhabited.

6.3 The five load-bearing pins

G1A-A: Poisson–Bruhat/source transport. After the one-sided physical transform and reciprocity, every retained source state ξ must have moving-prime coefficient

$$c_r(\xi) = B_\xi F_\xi(r/R), \quad (\text{G1A-PB})$$

where F_ξ is a uniformly smooth envelope generated from the one physical W_D , not an arbitrary row weight. The transport must preserve the physical/normalized energy up to a square-summable source error.

STATUS: SOURCE-OPEN / CONDITIONAL PIN.

Without this pin, the later moving-prime participation argument would apply to a convenient model rather than to the physical packet.

G1A-B: absolute source ceiling. The transported physical source must satisfy

$$\mathcal{T}_{\text{abs}} \ll M^2 L^4 X_1^{o(1)}. \quad (\text{G1A-abs})$$

This is the absolute comparator against which the moving-family contraction is measured.

STATUS: SOURCE/ANALYTIC PIN.

G1A-C: fixed-state exclusion and prime participation. A fixed nonprojective state must not absorb the moving prime variable. Smoothness of F_ξ produces a long plateau; the short-interval prime theorem then gives

$$N_\xi \gg R^{3/4-o(1)} \quad (\text{G1A-part})$$

participating prime rows for each retained state. The finite BPP compiler converts this participation into

$$E_{\text{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\text{abs}}. \quad (\text{G1A-np})$$

The project kernel-checks the finite participation-to-energy implication; it does not formalize the entire physical smooth-envelope/short-interval input.

Status. FINITE COMPILER LEAN-PROVED; the physical participation statement is EXTERNAL / SOURCE-OPEN.

G1A-D: outer-root physical energy and projective routing. The abstract nonprojective energy must be identified with the physical Gate energy, and the projective/axis/diagonal branch must be routed in the same source normalization. The finite ratio-class algebra includes the state diagonal in the zero-curvature branch, and the physical projective mass target is

$$E_{\text{proj}} \ll MHL^4 X_1^{o(1)}. \quad (\text{G1A-proj})$$

Exactly one outer square root converts (G1A-np) to an $R^{-1/4+o(1)}$ contraction. At the three controlling vertices the signed margins are

$$\frac{1}{72}, \quad \frac{1}{24}, \quad \frac{1}{32}. \quad (\text{G1A-margins})$$

Status. FINITE SQUARE-ROOT / PROJECTIVE ALGEBRA BANKED; the physical energy identification is SOURCE-OPEN.

G1A-E: recombination/discarded-sector bound. After all source truncations, exceptional routes and recombinations, the physical error must satisfy

$$E_{\text{rec}} \ll U^{-2+o(1)} M^2 L^4, \quad (\text{G1A-rec})$$

with the source-exact prefactor and without spending the same saving twice. The exponent margins at the controlling vertices are

$$\frac{1}{18}, \quad \frac{1}{18}, \quad \frac{1}{24}.$$

Status. EXPONENT BUDGET BANKED; literal source recombination is SOURCE-OPEN.

6.4 What the BPP mechanism proves and what it does not

The finite BPP mechanism is a local-to-family compiler:

$$\begin{array}{c} \text{smooth moving-prime participation} \\ \Downarrow \\ R^{-1/2+o(1)} \text{ family-energy contraction} \\ \Downarrow \\ R^{-1/4+o(1)} \text{ after one outer root.} \end{array}$$

It does *not* prove the source transport, absolute source ceiling, physical energy identification or recombination estimate. Those are separate statements precisely so that a reader can attack or falsify them independently.

6.5 The source-glue/SB- ν problem

The current source-glue question is whether the same physical packet can simultaneously satisfy the common-weight representation, the BPP state representation and the final routing representation with fibre multiplicity

$$N_\nu = X_1^{o(1)} \quad (\text{G1A-SBnu})$$

in the synchronized two-view model. This is a source-identification frontier rather than a new exponent computation.

Finally,

$$\text{GATE1A_REQUIRED_FOR_287} = \text{UNKNOWN.}$$

The Ford-generated packet census, not programme nomenclature, decides whether a needed #287 packet truly belongs to the canonical common-weight source class.

7 Gate 1B: signed/Möbius parity-breaking provider

7.1 Native centering, sign and complementary-divisor geometry

For coprime d, p ,

$$\rho_{dp}(y) = \rho_d(y)\rho_p(y) + \frac{\rho_d(y)}{p} + \frac{\rho_p(y)}{d}. \quad (\text{G1B-center})$$

If q is squarefree and $p \mid q$ is the distinguished prime, then

$$\mu(q/p) = -\mu(q). \quad (\text{G1B-sign})$$

This is the sign resource that must not be squared away accidentally.

The native complementary-divisor source exposes

$$\boxed{q\ell - uv = 2.} \quad (\text{G1B-det})$$

This determinant-two geometry is separate from the Gate-1A determinant $2k$ and from the #287 $\mu * \log$ determinant-one line.

7.2 The banked one-completion source

On dyadic windows $d \sim D$ and $p \sim P$, the authoritative one-completion coefficient is

$$\boxed{\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p),} \quad (\text{G1B-beta})$$

where μ_D, Λ_P are the physical truncated Möbius and prime weights on the stated windows. The public formal challenge proves the separate linearity/bilinearity of this coefficient and an exact level-of-distribution budget requiring

$$\boxed{\theta \geq \frac{1}{2} + \frac{1}{104}.} \quad (\text{G1B-104})$$

It also proves the implication from an inhabited source-to-convolution-BV interface to the abstract Gate conclusion. Producing that interface from the literal physical source remains open. This is a valid formal provider socket, but it is no longer the first research-level analytic obstruction.

7.3 The Motohashi line

Fix u, ℓ and a base solution to

$$\boxed{uv_0 + 2 = \ell z_0.} \quad (\text{G1B-base})$$

All solutions on the complementary-divisor line are

$$\boxed{v_t = v_0 + \ell t, \quad z_t = z_0 + ut.} \quad (\text{G1B-line})$$

Let H be the physical line length and write $e_H(x) = e^{2\pi i x/H}$. For the exact one-completion coefficient and a smooth cutoff W_1 define

$$\boxed{\mathcal{Z}_{u,\ell}(k) = \sum_t \beta_{D,P}(z_t) W_1(t/H) e_H(-kt).} \quad (\text{G1B-Z})$$

For the five centred defect factors

$$b_5 = \delta_1 * \delta_2 * \delta_3 * \delta_4 * \delta_5$$

and a second physical cutoff W_2 , define

$$\boxed{\mathcal{B}_{u,\ell}(k) = \sum_t b_5(v_t) W_2(t/H) e_H(-kt).} \quad (\text{G1B-B})$$

The Motohashi pairing is

$$\boxed{\mathcal{M}_{u,\ell} = \frac{1}{H} \sum_k \mathcal{Z}_{u,\ell}(k) \overline{\mathcal{B}_{u,\ell}(k)}.} \quad (\text{G1B-M})$$

The current supplied research ledger reports the exact A/B/C source pin, the twisted-defect A/B/C reduction and the fivefold Motohashi iteration as passed source algebra; the fixed-polylogarithmic- k interior is closed modulo the comparison convention.

7.4 Endpoint residue coefficient and diagonal/off-diagonal split

At the endpoint define, for a unit residue $a \bmod \ell$,

$$A_{\ell,a,k} = \sum_{u \equiv -2a^{-1} \pmod{\ell}} a_4(u) \mathcal{Z}_{u,\ell}(k), \quad (\text{G1B-A})$$

where a_4 is the four-defect coefficient inherited from the physical source. The endpoint Cauchy energy is

$$\mathcal{E}_{\text{end}}(k) = \sum_{\ell \sim L} \sum_{a \in (\mathbb{Z}/\ell\mathbb{Z})^\times} |A_{\ell,a,k}|^2. \quad (\text{G1B-Eend})$$

Opening the square gives

$$\mathcal{E}_{\text{end}}(k) = \sum_{\ell,a} \sum_{u_1, u_2 \equiv -2a^{-1} \pmod{\ell}} a_4(u_1) \overline{a_4(u_2)} \mathcal{Z}_{u_1,\ell}(k) \overline{\mathcal{Z}_{u_2,\ell}(k)}.$$

The diagonal is $u_1 = u_2$. The off-diagonal is parametrized by

$$u_2 = u_1 + j\ell, \quad j \neq 0. \quad (\text{G1B-j})$$

For corresponding line parameters t_1, t_2 ,

$$z_2(t_2) - z_1(t_1) = u(t_2 - t_1) + j v_{t_2}, \quad (\text{G1B-off-source})$$

and, after opening $\beta = \mu_D \Lambda_P$,

$$d_2 p_2 - d_1 p_1 = u(t_2 - t_1) + j v_{t_2}. \quad (\text{G1B-off-beta})$$

These identities are the literal arithmetic object behind the current endpoint off-diagonal problem.

The latest supplied research ledger records

$$\text{endpoint } u\text{-diagonal} = \text{power closed with } X^{-1/12+o(1)}. \quad (\text{G1B-diag-status})$$

7.5 Current first Gate-1B analytic theorem

Let $\mathfrak{T}_{1B}(X)$ denote the physical target scale of the Gate-1B parent after restoring the dyadic prefactor from the source decomposition. The current source materials provided to this editorial run specify the off-diagonal object (G1B-Eend)–(G1B-off-beta) but do not print the final numerical parent prefactor in a form that can be independently reconstructed here. We therefore do *not* invent it.

The current theorem slot is:

$$\sum_{\text{endpoint dyadic packets}} \mathcal{E}_{\text{end}}^{\text{off}} = o(\mathfrak{T}_{1B}(X)), \quad (\text{G1B-off-target})$$

with the exact physical prefactor/normalization to be copied verbatim from the controlling Gate-1B source before publication. Programme label:

$$\text{RANKONE-ENDPOINT-} \\ \text{U-OFFDIAG45}$$

Its source formula is exact; the missing printed normalization in this candidate is **SOURCE-BLOCKED NORMALIZATION**, and the cancellation itself is **ANALYTIC-OPEN**. This explicit split is preferable to guessing a power.

7.6 Downstream Gate-1B theorem objects

The first endpoint theorem is not the whole Gate. The current dependency chain is:

PURE5 comparison/main-term pin. The five-defect physical source and its comparison source must use the same principal, small-conductor and exceptional-character convention.

PURE5-COMPARISON-MAINTERM-PIN: SOURCE-OPEN.

Closing an oscillatory estimate with the wrong main term does not close the physical packet.

High- k sector. The same Motohashi pairing (G1B-M) restricted to the complementary high-frequency range defines RANKONE-HIGHK45. Its target is the Gate-1B parent bound on that frequency sector.

ANALYTIC-OPEN.

Pure-five signed distinguished-prime packet. After recombining interior, endpoint and high- k sectors, the source is the signed five-centred-defect distinguished-prime packet, schematically

$$\sum \mu_D(d) \Lambda_P(p) \prod_{i=1}^5 \delta_i(\cdot) \times (\text{physical cutoffs}),$$

with all congruence and gcd restrictions inherited from (G1B-det). The literal final prefactor is part of the source packet.

PURE5-DP-SIGNED45: DOWNSTREAM ANALYTIC/REASSEMBLY OPEN.

Lower defect orders. The $|J| = 4, 3, 2, 1$ packets are separate centred-defect strata obtained by selecting which defect factors remain nontrivial. They do not follow automatically from the Pure5 theorem; each requires its own source routing or a proved monotonicity/reduction theorem.

Near-primitive and conductor complement. The near-primitive $r = 1$ full-conductor slice is denoted NEARPRIM-DP-SIGNED45. The $r > 1$ family, the (C_*, C_{nw}) transition, and proper-divisor recursion must then restore the remaining conductor ranges without losing the Möbius sign.

QK56 parent. The full covariance parent has the schematic but literal two-modulus form

$$\mathcal{C}_{\text{QK}} = \sum_{q_1, q_2} \sum_{h_1, k_1, h_2, k_2} A(q_1, q_2; h_1, k_1, h_2, k_2) Q(q_1; h_1, k_1) \overline{Q(q_2; h_2, k_2)}. \quad (\text{QK56-parent})$$

The branch $q_1 = q_2$ is the same-modulus character Gram; $q_1 \neq q_2$ is the cross-modulus/CRT branch. The canonical zero mode and substantial finite/source decompositions are banked, but QK56-EXHAUSTIVENESS must prove that all physical children are covered. QK56 is downstream of the rank-one endpoint in the current closure order, not the present first analytic open.

Shifted TT^* and finite reassembly. The final shifted source must be recovered before any support enlargement; all zero/nonunit, collision and repeated-prime strata must be routed; then the finite source compiler reassembles the leaves into the Gate-1B parent.

Only after these steps may one write GATE1B CLOSED.

8 Ford–Maynard generated packets and proof-specific Type II

8.1 Published universal wrapper

The Ford–Maynard lower-bound framework uses a universal Type-II hypothesis. In the present 1/6 specialization a safe sufficient form is

$$\left| \sum_{\substack{(X/2)^\varepsilon < m \leq X^{1/6-\varepsilon} \\ X/2 < mn \leq X}} \xi_m \kappa_n w_X(mn) \right| \ll_A X (\log X)^{-A} \quad (\text{FM-universal-II})$$

for every allowed divisor-bounded coefficient pair. The project does *not* claim that Ford–Maynard publish the weaker proof-specific theorem described next.

8.2 What their proof actually generates

The Ford arithmetic source has

$$G(m; n) = \mathbf{1}_{m \leq n^\gamma} \lambda(m_1) g(\mathbf{v}(m_2; n)), \quad H(n) = \sum_{d|n} G(d; n), \quad (\text{FM-GH})$$

with the canonical rough/smooth split $m = m_1 m_2$. After finite decomposition and condition removal, Proposition 7.22 reaches a fixed-depth product sum (equation (7.23) in the source)

$$\sum_{n=n_1 \cdots n_N \asymp x} w_n x_1(n_1) \cdots x_N(n_N), \quad (\text{FM-7.23})$$

and chooses a product $\prod_{e \in E} n_e$ in the Type-II range. Grouping the selected factors and their complement gives

$$\sum_{n=n' n'' \asymp x} Y_1(n') Y_2(n'') w_{n' n''}. \quad (\text{FM-generated-II})$$

The published proof then invokes the universal Type-II hypothesis.

Hence an invocation-by-invocation replacement theorem would be enough for this proof:

$$\begin{array}{c} \text{every actual generated (FM-generated-II) packet} \\ \text{has a proved source-exact provider} \\ \Downarrow \\ \text{every Type-II invocation in Proposition 7.22 is covered} \end{array} \quad (\text{FM-replacement-logic})$$

This implication additionally requires a source-exhaustive census and a replacement certificate; it is not obtained merely by identifying some packets.

8.3 The three source obligations

Definition 8.1 (SOURCE-ADAPTER45). For every generated packet, print the literal operator, physical ranges, coefficient class, weight dependence, centering/expected term, gcd/collision restrictions, original prefactor and normalized target, and map it to a proved provider without changing the source class.

Definition 8.2 (FM-TO-GATE-COORDINATE-CENSUS45). Enumerate every Type-II packet actually generated by the Ford decomposition, with enough coordinate information to decide whether it belongs to Gate 1A, Gate 1B or a direct exceptional/standard class.

Definition 8.3 (C_{FM} replacement certificate). Verify invocation by invocation that the generated census covers every use of the universal Type-II hypothesis in the chosen Ford proof, with no omitted packet, no multiplicity loss and no double-spending of an analytic saving.

All three remain open source/reassembly obligations.

8.4 Provider-facing generated families

Family	Literal source/scale	Candidate provider	Why it matches / what is missing
Möbius singleton	one selected $m < X^{1/6}$ fac- tor carries Möbius weight; complement remains generated	Gate 1B signed source	Möbius sign is in the cor- rect physical slot only after a literal completion dictionary; generated outer-factor stabil- ity is still required.
Model singleton	selected factor is model/Mellin-like; Möbius carrier may remain in the complement	QK56 or direct affine completion	provider cannot be inferred from the word “singleton”; na- tive packet dictionary is open.
Balanced-seven V- cell	seven fixed prime boxes at $Y = X^{1/7}$	factorial- Ω_7 source, then the fixed- degree-seven attack	factorial Euler encoding is ex- act; endpoint/3221 analytic theorem and comparison re- main.
Pure5/high-order family	five centred de- fect factors with distinguished- prime/Möbius source	Gate 1B Motohashi/rank- one chain	current first analytic object is the endpoint off-diagonal; high- k and downstream de- fect/conductor families re- main.
Balanced R_9	nine-coordinate Ford-generated leakage cell	fixed- certificate/FCL lane	finite binomial 70 is proved; literal physical $G(d; n)$ special- ization is source-open.
Common-weight packets	actual generated source must pre- serve one physical W_D after scale conversion	Gate 1A	requires exact scale/source dic- tionary and all five Gate-1A pins.
Remaining fixed- g_* packets	not yet exhaustively censused	TBD from literal source	the finite census/provider as- signment itself is still open.

9 Balanced R_9 , fixed-certificate leakage and FCL

The balanced order-nine finite identity is

$$\boxed{\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70.} \quad (\text{R9-70})$$

This is PROVED-FINITE; by itself it is not the physical Ford statement $H(n) = 70$.

To obtain $H(n) = 70$ one must prove that the literal Ford specialization of $G(d; n)$ on the balanced R_9 cell selects exactly the corresponding low/high divisor orders and has the required value of $g(\emptyset)$. That is

$$\boxed{\text{literal } G(d; n) \text{ specialization: SOURCE-OPEN.}} \quad (\text{R9-G-open})$$

Therefore balanced R_9 is a *Ford-generated leakage candidate*, not a completed leakage theorem.

For one fixed positive Ford certificate g_* , define its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Let $\mathcal{P}$ be the desired prime region, $\mathcal{N}_1$ the certificate-sign composite region, $\mathcal{N}_2$ the exceptional collar, and

$$\mathcal{U}_* = (X/2, X] \setminus (\mathcal{P} \cup \mathcal{N}_1 \cup \mathcal{N}_2).$$

The fixed-certificate leakage is

$$\boxed{\mathcal{L}_*(X) = \sum_{n \in \mathcal{U}_*} w_X(n) H_*(n).} \quad (\text{FCL})$$

The N_2 error is

$$\mathcal{E}_{N_2}(X) = \sum_{n \in \mathcal{N}_2} w_X(n) H_*(n).$$

If the total correlation, leakage, N_2 error and comparison error are all small relative to the positive certificate margin $C_*(\varepsilon_*)$, then the fixed-certificate transference gives positive affine prime mass. A sufficient quantitative form is

$$|\mathcal{L}_*(X)| + |\mathcal{E}_{N_2}(X)| \leq \frac{C_*(\varepsilon_*)}{4} B_X, \quad (\text{FCL-margin})$$

together with the corresponding total-correlation/comparison bounds. This is a conditional compiler; it does not itself prove FCL.

Balanced R_9 may contribute to $\mathcal{L}_*$ once the source obligation (R9-G-open) is discharged, but closing that one cell does not close the full leakage family.

10 Gate 2: source-specific Ford–Maynard lower-bound compiler

Fix one compatible

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Gate 2 is the theorem-shaped endgame package:

Theorem 10.1 (Gate-2 lower-bound compiler – conditional). *Assume:*

G2.1 *the Gate-0 source/comparison/Type-I package;*

G2.2 *the literal Ford $G(d; n)/H(n)$ source and its sign properties;*

G2.3 *a source-specific aggregate N_2 estimate;*

G2.4 *one common fixed $\varepsilon_* > 0$ compatible with every source range;*

G2.5 *either the universal Type-II theorem (FM-universal-II), or a complete generated-packet provider package together with **SOURCE-ADAPTER45**, **FM-TO-GATE-COORDINATE-CENSUS45** and C_{FM} ;*

G2.6 *the ordinary/bounded-class firewall and every comparison/exceptional-character normalization needed by the chosen Ford theorem;*

G2.7 *effective versions of all inputs at a computable threshold.*

Then the Ford–Maynard lower-bound geometry yields positive affine mass of the form

$$\boxed{\sum_{p \in \mathcal{P}} a_X(p) \geq (C_*(\varepsilon_*) + o(1)) B_X \gg \frac{X}{\log X},} \quad (\text{G2-positive})$$

for the physical #287 source and the chosen certificate.

STATUS: CONDITIONAL-COMPILER / NOT ACTIVATED.

The theorem still does not by itself produce **WindowPairSupply**. One additional effective witness extraction theorem must convert the positive affine mass into a Sophie/safe-prime or prime-power blocker configuration, or directly into the **WindowPairSupply**(M) predicate of the finite compiler. That final conversion/effectivity step remains open.

11 Current #287 specialization: $\mu * \log$, factorial Ω_7 , and the 3221 attack

The source-minimal identity used in the current #287 branch is

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r, \quad s = \pm 1. \quad (287\text{-mulog})$$

The latest supplied V15 replay reports the exact three-way split into small- q , small- r , and hard pairs as kernel-checked, together with the determinant-one line

$$qr - 2mn = s.$$

For fixed r, m and one solution,

$$q = q_0 + 2mt, \quad n = n_0 + rt.$$

11.1 Factorial Euler seven-box source

For the balanced-seven cell

$$Q = X^{3/5}, \quad Y = X^{1/7},$$

let

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p), \quad F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}. \quad (\text{Omega7-factorial})$$

Then the degree- $(1, \dots, 1)$ torus coefficient exactly recovers the ordered seven-labelled prime source, including repeated primes:

$$7^7 [z_1 \cdots z_7] F_{\mathbf{z}}(n) = \sum_{\substack{p_1 \cdots p_7 = n \\ p_i \text{ ordered}}} \prod_{i=1}^7 \omega_i(p_i). \quad (\text{Omega7-coeff})$$

The latest supplied V16 factorial replay reports the general/seven-slot polarization, repeated-prime fibre cardinality, formal local Euler algebra and expected-term linearity as Lean-proved in an 8123-job build.

The exact balanced-seven source therefore reduces to

$$\int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, -s\overline{2}) dz$$

plus the physical comparison identification. The exact broad endpoint theorem is

$$\text{AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45} \quad (\text{Omega7-endpoint})$$

It is strictly narrower than a BV theorem for all class- $\mathcal{C}$ functions, an absolute modulus average, a theorem uniform in every $\mathbf{z}$, or a general smooth-number theorem.

11.2 Why the direct existing Pascadi proof route does not splice

In the audited current v2 proof architecture, a small parameter η is arranged so that

$$Q \leq x^{5/8-100\eta}, \quad y \leq x^{\eta}.$$

At $Q = x^{3/5}$,

$$\eta \leq \frac{1}{4000},$$

whereas $y = x^{1/7}$ requires

$$\eta \geq \frac{1}{7}.$$

Hence the direct written proof route is nonclosing at the required smoothness scale. This is a parameter/dictionary audit, not a theorem ruling out future extensions.

11.3 The current 1+2+2+2 / 3221 research reduction

The latest supplied Pro reduction keeps the source at fixed degree seven and groups the seven prime boxes as

$$\boxed{1 + 2 + 2 + 2.}$$

Use the scale ledger

$$\boxed{E = X^{1/7}, \quad M = N = L = X^{2/7}, \quad Q = X^{3/5}.} \quad (3221\text{-scales})$$

The ordinary three-block comparison sees

$$\frac{4}{7} - \frac{3}{5} = -\frac{1}{35},$$

so the naive $Q < NL$ geometry misses by $X^{1/35}$. Retaining the actual seventh box inside Cauchy changes the controlling comparison to

$$\boxed{Q < ENL}$$

with exponent margin

$$\boxed{\frac{5}{7} - \frac{3}{5} = \frac{4}{35}.} \quad (3221\text{-margin})$$

The source-assisted diagonal audit gives an amplitude saving

$$\boxed{X^{-2/35+o(1)}} \quad (3221\text{-saving})$$

After opening the two copies, the literal difference relation is

$$\boxed{e_1 n_1 \ell_1 - e_2 n_2 \ell_2 = qt, \quad |t| \ll X^{4/35+o(1)}} \quad (3221\text{-diff})$$

The line-length ledger is

$$\boxed{H = X^{11/35}, \quad EH = X^{16/35} < Q.} \quad (3221\text{-H})$$

The current proposed narrower source socket is

$$\boxed{\begin{array}{c} 3221\text{-DI-KUZNETSOV-} \\ \text{LITERAL-SPLICE45} \end{array}} \quad (3221\text{-label})$$

Its theorem is not “prove Kuznetsov”. It is the source-dictionary/analytic statement: after the above post-dispersion decomposition, identify the literal off-diagonal (3221-diff) with a DI/Kuznetsov-type bilinear form *without changing the physical weights, Möbius sign, prefactor or multiplicity*, and obtain a fixed-power saving sufficient to recover at least the source-assisted margin (3221-saving). If such a literal splice is proved, it is a narrower sufficient route toward (Omega7-endpoint).

STATUS: RESEARCH CANDIDATE / SOURCE-DICTIONARY + ANALYTIC OPEN.

The supplied materials do not contain a completed hostile audit of this splice, so it is not promoted above the exact factorial endpoint theorem.

12 Verification manual: what is actually proved

12.1 Table 1: machine-checked objects

Object / declaration	File or replay	Exact content	Does not prove
no_Erdos287 Counterexample_ of_max_le_4e9	FiniteRangeExtension. lean	No exact counterexample with maximum denominator at most $4 \cdot 10^9$.	No large- M theorem.
WindowPairSupply; no_Erdos287 Counterexample_ of_closure vaughan_identity_ exact; source compilers	ClosureInputs.lean / end-to-end status bank	Exact conditional compiler from effective eventual supply plus finite range.	No inhabitant of eventual supply.
	V14 affine Vaughan files	Exact alternative source decomposition and structural algebra.	No Type-II estimate.
vonMangoldt_eq_ mobius_mul_log; muLog_affine_ pointwise three-way pair partition / muLogHard	latest supplied V15 replay: AffineMuLogIdentity. lean	$\Lambda = \mu * \log$ and its affine specialization.	No cancellation.
determinant-one line family	latest supplied V15 replay: AffineMuLogLine.lean	Exact disjoint small- q /small- r /hard decomposition.	No smallness of any piece.
		Coprimality, affine parametrization and multiplicity-one uniqueness.	No dispersion estimate.
balanced-seven –20	BalancedSevenFinite. lean in latest supplied V15 replay	Finite alternating-binomial coefficient.	No physical Ford source specialization.
squarefree labelled po- larization	latest supplied V15 replay: BalancedSeven Polarization.lean	Exact squarefree coefficient extraction and normalization.	Superseded as controlling by factorial encoding.
OMEGA7-FACTORIAL- EULER- POLARIZATION45	latest supplied V16 factorial replay; exact public filename synchronization pending	General/seven-slot factorial polarization, including repeated-prime fibre accounting.	No endpoint distribution estimate.
POLARIZED- EXPECTED-TERM- LINEARITY45	latest supplied V16 factorial replay	Coefficient extraction commutes with every linear expected-term operator.	No physical identification $\mathfrak{M}^{\text{fac}} = \mathfrak{M}^{\text{phys}}$.
Pascadi (3/5, 1/7) pa- rameter ledger	latest supplied V16 factorial replay	Formal rational inequality $\eta \leq 1/4000$ versus $\eta \geq 1/7$.	No theorem against future analytic extensions.

The supplied V16 factorial report states `lake build`: 8123 jobs, zero errors; no Lean-code occurrences of `sorry`, `admit`, user `axiom`, `opaque`, `unsafe`, `native_decide` or `@[implemented_by]`; principal axiom reports use only the standard logical principles `propext`, `Classical.choice` and `Quot.sound`. These are replay-status claims supplied with the project; public aggregate synchronization is tracked separately.

12.2 Table 2: externally audited / mathematical inputs

Input	Status	Role / limitation
Input	Status	Role / limitation
Huxley short-interval prime theorem	EXTERNALLY-AUDITED	Supplies prime participation once the Gate-1A physical smooth envelope is source-pinned.
Ford–Maynard lower-bound framework	EXTERNALLY-AUDITED	Publishes the universal Type-II sufficient wrapper and lower-bound geometry; does not publish the project’s proof-specific replacement theorem.

Drappeau–Granville–Shao class- $\mathcal{C}$ distribution	EXTERNALLY-AUDITED / RANGE MISMATCH	Relevant multiplicative-function technology; does not literally give the exact $Q = X^{3/5}$ endpoint at the required source.
Pascadi smooth-number proof architecture	EXTERNALLY-AUDITED / DIRECT ROUTE NONCLOSING	At $Q = X^{3/5}$ its written parameter architecture forces $\eta \leq 1/4000$, incompatible with $y = X^{1/7}$.
Latest Motohashi/rank-one Gate-1B reductions	RESEARCH-AUDITED, NOT PUBLIC THEOREMS	Source reductions supplied after the V15 snapshot; first analytic object is the endpoint off-diagonal, with exact parent normalization still to be copied into the publication draft.
Latest 3221 reduction	RESEARCH CANDIDATE	Narrows the factorial endpoint to a post-dispersion fixed-degree-seven source socket if the DI/Kuznetsov literal splice survives hostile audit.

12.3 Table 3: open source identifications

Source object	Status	What must be printed/proved
Source object	Status	What must be printed/proved
Gate-0 comparison $(b.2)/$ Type-I normalization	SOURCE-OPEN	Uniform physical progression/coagulation convention and literal weighted/maximal theorem.
Gate-1A PB/source glue and physical energy pin	SOURCE-OPEN	Exact transport, common-weight preservation, absolute ceiling, participation state, energy identification, projective routing and recombination.
Gate-1B endpoint physical target prefactor	SOURCE-BLOCKED NORMALIZATION	Copy the literal parent prefactor/target accompanying (G1B-off-target); do not infer it from the programme label.
PURE5-COMPARISON-MAINTERM-PIN	SOURCE-OPEN	Principal/small-conductor/exceptional-character physical main-term convention.
Ford generated-(7.23) census	SOURCE-OPEN	Every actual packet, scale, coefficient class, weight law, multiplicity and provider.
SOURCE-ADAPTER45 / C_{FM}	SOURCE-OPEN	Literal Gate/provider dictionary and invocation-by-invocation replacement certificate.
Balanced- R_9 physical $G(d; n)$ specialization	SOURCE-OPEN	Show that the physical Ford cell really realizes the divisor-order split leading to 70.
Factorial/physical comparison	SOURCE-OPEN	Prove $\mathfrak{M}_{7,s}^{\text{fac}} = \mathfrak{M}_{7,s}^{\text{phys}}$ in the exact local-density and exceptional-character convention.
3221 DI/Kuznetsov completed-source dictionary	SOURCE-OPEN	Identify the post-dispersion source with the intended theorem socket with no loss of sign, weights, prefactor or multiplicity.

12.4 Table 4: open analytic theorems in current dependency order

Open theorem	Exact task	What remains even if proved
Open theorem	Exact task	What remains even if proved

3221-DI-KUZNETSOV-LITERAL-SPLICE45	Prove the literal post-dispersion 1+2+2+2 source admits a DI/Kuznetsov-type estimate with a fixed-power saving sufficient for the source-assisted margin (3221-saving).	Physical comparison and the rest of the factorial endpoint/source census; status remains a research candidate until the source dictionary is hostile-audited.
AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45	Bound the signed degree-(1, ..., 1) torus coefficient of the $F_{\mathbf{z}}$ discrepancy by $o(X/\log X)$.	Physical comparison match; other V-cells and M-branch.
RANKONE-ENDPOINT-U-OFFDIAG45	Bound the off-diagonal endpoint energy (G1B-Eend)–(G1B-off-beta) relative to the physical Gate-1B parent target.	Pure5 comparison, high- k , lower defects, conductor recursion, QK56 and reassembly.
RANKONE-HIGHK45	Control the high-frequency part of (G1B-M) in the physical source normalization.	Pure5/lower-defect/conductor/QK56 chain.
Remaining generated packet providers	Prove each uncensused Ford-generated packet has a valid Gate 1A, Gate 1B or direct provider.	Replacement certificate, FCL, Gate 2.
FCL	Control the leakage sum (FCL) and the associated total-correlation/ N_2 /comparison errors within the fixed certificate margin.	Gate 2 source package/effectivity and large- M witness extraction.
N_2 /Gate-2 analytic inputs	Prove the source-specific aggregate exceptional-collar estimate and all remaining lower-bound hypotheses at one ε_* .	Positive mass still must be converted effectively to WindowPairSupply.
eventual WindowPairSupply	Prove WindowPairSupply(M) for every sufficiently large M , or prove a stronger effective witness implying it.	Nothing analytic: the existing finite Lean compiler then gives Erdős #287.

13 Current numbered route to Erdős #287

The current literal route, subject to future simplification, is:

1. finish or supersede the current fixed-degree-seven/3221 socket;
2. prove the factorial physical comparison identification;
3. complete the remaining balanced-seven/V-cell census;
4. control the Möbius singleton and model singleton packets;
5. control the $k = 0$ smooth-parity family;
6. complete the remaining fixed- g_* packet census;
7. assign every generated packet to a proved Gate 1A, Gate 1B or direct provider;
8. complete the generated-(7.23) replacement certificate C_{FM} ;
9. close the full fixed-certificate leakage/correlation package;
10. prove the source-specific N_2 estimate;
11. activate Gate 2 at one compatible ε_* and obtain positive affine mass;
12. make the lower-bound/witness extraction effective;
13. prove eventual WindowPairSupply(M) (or a stronger effective witness);
14. invoke the existing finite Lean compiler.

A future theorem may collapse several steps; the list is a current dependency audit, not a claim of logical minimality.

14 How an external reviewer can falsify or improve the programme

A serious review is most useful if it attacks one of the following:

1. a source dictionary that changes the physical packet;
2. a theorem-slot mismatch between an actual generated packet and a claimed provider;
3. a missing normalization or prefactor;
4. a multiplicity or support-enlargement loss;

5. double-spending of a Möbius sign or analytic saving;
6. an incorrect comparison/principal/exceptional-character main term;
7. an incomplete Ford-generated packet census;
8. an invalid invocation-by-invocation replacement of Ford–Maynard’s universal Type II;
9. an endpoint/effectivity gap;
10. a mismatch between a claimed Lean status and the actual kernel/public bank.

Failure of any conditional provider does *not* invalidate the machine-checked finite theorem through 4×10^9 ; it invalidates only the corresponding large- M route.

15 Glossary of recurring programme terms

Gate 0	The physical #287 source/comparison/Type-I package.
Gate 1A	Canonical common-weight moving-source Type-II provider (G1A-source).
Gate 1B	Complementary signed/Möbius/high-order Type-II provider based on (G1B-center)–(G1B-M).
Gate 2	Source-specific Ford–Maynard lower-bound compiler.
BPP	Band-limited prime-participation mechanism converting moving-prime participation to a family-energy saving in Gate 1A.
SB-ν	Synchronized two-view/source-glue fibre condition in the Gate-1A source adapter.
Pure5	Five-centred-defect distinguished-prime source family in the Gate-1B chain.
NearPrim	Near-primitive/full-conductor slice ($r = 1$) of the Gate-1B conductor family.
QK56	Full same-/cross-modulus covariance parent (QK56-parent).
SOURCE-ADAPTER45	Literal source dictionary from actual Ford-generated packets to proved providers.
C_{FM}	Invocation-by-invocation certificate replacing the universal Type-II hypothesis in the chosen Ford proof by generated-packet estimates.
R_9	Balanced nine-coordinate Ford-generated leakage candidate; finite binomial coefficient 70 is proved, physical $G(d; n)$ specialization is not.
FCL	Fixed-certificate leakage/correlation package (FCL).
N_2	Exceptional Ford collar requiring a separate source-specific aggregate bound.
Balanced7	Fixed-degree seven-prime-box V-cell at $Y = X^{1/7}$.
V-cell	A generated packet in the model/factorial seven-prime branch.
Möbius singleton	Generated short Type-II factor carrying the Möbius sign.
fixed-g_*	One fixed positive Ford certificate used in the leakage transference.
WindowPairSupply	The exact large- M adjacent prime-power window predicate consumed by the finite Lean compiler.

16 The source-minimal affine algebra

The current clean narrative starts from the exact identity $\Lambda = \mu * \log$. The supplied V15 Aristotle replay report records a successful 8118-job kernel build of the declarations summarized below. Because public repository **main** has not yet synchronized that replay, their status in this candidate is **LEAN-PROVED IN LATEST V15 REPLAY**, not **PUBLIC LEAN-CHECKED**. No analytic estimate is encoded by these identities.

16.1 The identity $\Lambda = \mu * \log$

For the genuine arithmetic functions,

$$\mu * \log = \mu * (\Lambda * 1) = (\mu * 1) * \Lambda = \Lambda,$$

and hence

$$\Lambda(N) = \sum_{qr=N} \mu(q) \log r.$$

At the affine value $N = 2mn + s$, $s \in \{-1, +1\}$,

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r.$$

The V15 replay also preserves the natural-number sign firewall for the minus case.

16.2 Exact factor partition and determinant-one line

For a cutoff U , factor pairs split disjointly into

$$q \leq U, \quad q > U, r \leq U, \quad q > U, r > U.$$

The hard class is exact rather than heuristic:

$$\sum_{\substack{qr=2mn+s \\ q>U, r>U}} \mu(q) \log r.$$

Every hard pair satisfies

$$qr - 2mn = s, \quad s = \pm 1.$$

This implies

$$(r, 2m) = 1, \quad (q, 2n) = 1.$$

Fixing r, m and one solution (q_0, n_0) , all integer solutions are

$$\boxed{q = q_0 + 2mt, \quad n = n_0 + rt, \quad t \in \mathbb{Z},}$$

with unique t . The V15 replay report records nonvanishing, the coprimality statements, both directions of this parametrization, and multiplicity-one uniqueness as kernel-checked.

The same replay proves only the finite exponent ledger, not an asymptotic estimate. In particular,

$$\frac{1}{6} - \frac{16623}{100000} = \frac{131}{300000} > 0.$$

17 The balanced-seven factorial Euler reduction

Freeze the literal balanced-seven source

$$Q = X^{3/5}, \quad Y = X^{1/7},$$

and

$$\mathcal{B}_{7,s}(X) = \sum_{q \sim Q} \mu(q) \left[\sum_{\substack{p_1, \dots, p_7 \sim Y \\ 2p_1 \cdots p_7 \equiv -s \pmod{q}}} \prod_{i=1}^7 \omega_i(p_i) W(p_1 \cdots p_7 / X) - \mathfrak{M}_{7,s}^{\text{phys}}(q) \right].$$

For odd q , write

$$a_s(q) \equiv -s 2^{-1} \pmod{q}.$$

The target is $o(X/\log X)$. The V15 Lean replay already proves the finite coefficient

$$\sum_{j=0}^3 (-1)^j \binom{7}{j} = -20$$

and the earlier squarefree labelled polarization. The controlling representation below is newer and removes the collision remainder entirely.

17.1 Factorial Euler polarization

For $\mathbf{z} = (z_1, \dots, z_7) \in \mathbb{T}^7$, define

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p)$$

for primes in the actual box, and $a_{\mathbf{z}}(p) = 0$ outside it. Define a multiplicative function $F_{\mathbf{z}}$ by the prime-power data

$$\boxed{F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}, \quad e \geq 0.}$$

This is ordinary multiplicativity; complete multiplicativity is neither required nor desirable here.

Its local Dirichlet factor is

$$\sum_{e \geq 0} F_{\mathbf{z}}(p^e) p^{-es} = \exp(a_{\mathbf{z}}(p) p^{-s}),$$

so the generalized von Mangoldt coefficients satisfy

$$\Lambda_{F_{\mathbf{z}}}(p) = a_{\mathbf{z}}(p) \log p, \quad \Lambda_{F_{\mathbf{z}}}(p^e) = 0 \quad (e \geq 2).$$

Since $|a_{\mathbf{z}}(p)| \leq 1$,

$$|\Lambda_{F_{\mathbf{z}}}(n)| \leq \Lambda(n),$$

so $F_{\mathbf{z}}$ lies in the multiplicative class $\mathcal{C}$ used by Drapeau–Granville–Shao. This is a research/algebraic derivation; the factorial polarization has not yet completed a Lean replay.

17.2 Exact labelled coefficient extraction, including repeated primes

Let

$$n = \prod_p p^{e_p}, \quad \Omega(n) = \sum_p e_p = 7.$$

Then

$$F_{\mathbf{z}}(n) = \prod_p \frac{1}{e_p!} \left(\frac{1}{7} \sum_i z_i \omega_i(p) \right)^{e_p}.$$

The factors $e_p!$ cancel exactly the multiplicity from permuting equal prime occurrences. Therefore

$$7^7 [z_1 \cdots z_7] F_{\mathbf{z}}(n) = \sum_{\substack{p_1 \cdots p_7 = n \\ p_i \text{ ordered}}} \prod_{i=1}^7 \omega_i(p_i).$$

Consequently,

$$\begin{aligned} & \sum_{\substack{p_1, \dots, p_7 \sim Y \\ 2p_1 \cdots p_7 \equiv -s \pmod{q}}} \prod_{i=1}^7 \omega_i(p_i) W(p_1 \cdots p_7 / X) \\ &= 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{\substack{n \asymp X \\ n \equiv a_s(q) \pmod{q}}} F_{\mathbf{z}}(n) W(n/X) d\mathbf{z}. \end{aligned}$$

This identity is exact for repeated primes. Hence the former $O(X^{6/7+o(1)})$ repeated-prime correction is retired from the controlling proof architecture. The earlier collision router remains a valid historical check in the technical dossier.

17.3 Expected-term linearity and the exact comparison pin

Let $\mathcal{E}_q(f; a)$ be any expected-term operator that is linear in f : principal projection, small-conductor subtraction, exceptional-character contribution, or the literal combination adopted by the eventual analytic theorem. Linearity gives

$$7^7 [z_1 \cdots z_7] \mathcal{E}_q(F_{\mathbf{z}}; a) = \mathcal{E}_q(7^7 [z_1 \cdots z_7] F_{\mathbf{z}}; a).$$

Thus polarization itself creates no comparison obstruction.

Define

$$\mathfrak{M}_{7,s}^{\text{fac}}(q) = 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \mathcal{E}_q(F_{\mathbf{z}}; a_s(q)) d\mathbf{z}$$

and

$$\Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, a) = \sum_{\substack{n \asymp X \\ n \equiv a \pmod{q}}} F_{\mathbf{z}}(n) W(n/X) - \mathcal{E}_q(F_{\mathbf{z}}; a).$$

Then

$$\begin{aligned} \mathcal{B}_{7,s}(X) &= 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, a_s(q)) d\mathbf{z} \\ &\quad + \sum_{q \sim Q} \mu(q) (\mathfrak{M}_{7,s}^{\text{fac}}(q) - \mathfrak{M}_{7,s}^{\text{phys}}(q)). \end{aligned}$$

There is no repeated-prime remainder.

The comparison problem is therefore not “does polarization commute with the main term?” It does. The remaining source question is the physical identification

$$\mathfrak{M}_{7,s}^{\text{fac}}(q) = \mathfrak{M}_{7,s}^{\text{phys}}(q)$$

at the required aggregate precision: principal local density together with the literal exceptional-character convention. This remains AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45; it is not closed.

17.4 The first exact V-branch analytic residual

The controlling analytic statement is now

$$\int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, a_s(q)) d\mathbf{z} = o(X/\log X).$$

Programme label:

AFFINE287-FACTORIAL-OMEGA7-
SIGNED-ENDPOINT45

It is deliberately narrower than each of the following:

- a Bombieri–Vinogradov theorem for every $f \in \mathcal{C}$;
- an absolute average $\sum_q |\Delta_q|$;
- a theorem uniform in every $\mathbf{z} \in \mathbb{T}^7$;
- a general smooth-number distribution theorem.

Only the signed $\mu(q)$ -average of one degree- $(1, \dots, 1)$ torus coefficient is needed.

Together,

AFFINE287-FACTORIAL-OMEGA7-
SIGNED-ENDPOINT45
+ AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45
 $\Downarrow$
AFFINE287-BALANCED7-MODULUS-AVERAGE45 CLOSED.

Both antecedents remain open.

18 Published distribution results and the exact dictionary audit

Drappeau–Granville–Shao work with the class $\mathcal{C}$ of *multiplicative* functions satisfying $|\Lambda_f(n)| \leq \Lambda(n)$; complete multiplicativity is only a special case. Their theorem reaches moduli $x^{3/5-\varepsilon}$ for sufficiently smooth support, so it does not literally supply the exact endpoint $Q = X^{3/5}$ required above.

For Pascadi’s current arXiv v2, the relevant stated result is the smooth-number theorem (Theorem 1.5, refined as Theorem 6.4). Immediately after Theorem 1.5 the paper remarks that a similar result for smooth-supported multiplicative functions can be deduced by a slight extension of Proposition 6.3; the current v2 does not state a separate “Theorem 1.7” giving the black box required here. More importantly, the written proof architecture itself is quantitatively nonclosing at the seven-prime smoothness scale.

Indeed, renaming the proof’s internal small parameter as η , the proof arranges

$$Q \leq x^{5/8-100\eta}, \quad y \leq x^\eta.$$

At the required modulus

$$Q = x^{3/5},$$

the first inequality implies

$$\frac{3}{5} \leq \frac{5}{8} - 100\eta \implies \boxed{\eta \leq \frac{1}{4000}}.$$

But the source has

$$y = x^{1/7},$$

so the second inequality would require

$$\boxed{\eta \geq \frac{1}{7}}.$$

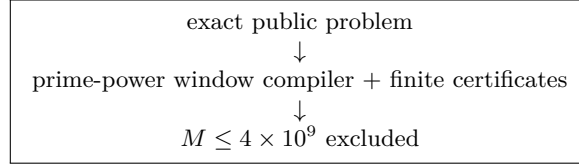
These conditions are incompatible. Thus the *direct existing Pascadi proof route*, with its written parameters, does not furnish the factorial endpoint theorem at $(Q, y) = (X^{3/5}, X^{1/7})$. This is a theorem-dictionary/proof-parameter audit, not a claim that no future extension of Pascadi’s technology can succeed.

This explains the current research direction: preserve the literal fixed degree seven rather than forcing a generic smooth-factorization parameter, carry that structure through dispersion, and test a post-dispersion four-block arrangement. The “fixed-degree-seven / four-block” programme is a research direction, not a proved theorem or an already matched Gate-1B provider. Its purpose is precisely to attack the narrower signed torus coefficient without paying for a theorem uniform over all smooth-supported functions.

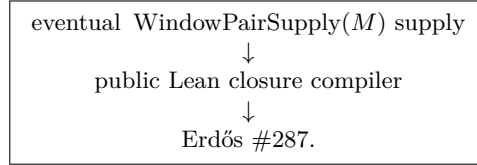
The earlier V14 Vaughan decomposition remains mathematically valid as a nonminimal alternative source audit. It is no longer presented as a co-equal controlling branch in this public narrative; its prime-modulus two-outer packet and related failed dictionaries belong in the technical provenance appendix.

19 The current end-to-end dependency graph

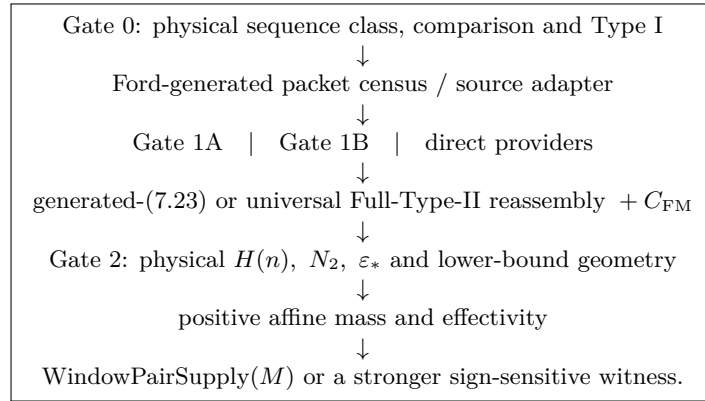
The unconditional finite theorem and the conditional large- M interface remain the two endpoints of the project:



while the conditional completion is



Between them sits the restored analytic gate architecture:



The current 287 overlays enter this spine at the generated-packet level:

$$\Lambda = \mu * \log, \quad F_{\mathbf{z}}(p^e) = a_{\mathbf{z}}(p)^e / e!, \quad \text{FACTORIAL-OMEGA7-SIGNED-ENDPOINT45},$$

together with the physical comparison identification. Even if the balanced-seven pair of residuals is closed, one still needs the remaining V-cells/M-branch, the fixed- g_* packet census, provider assignment, balanced R_9 source specialization where relevant, FCL/leakage reassembly, N_2 , the Gate-2 lower bound and an effective large- M witness. This is why the clean paper retains the Gate spine rather than compressing every analytic arrow into the phrase “hard cancellation”.

20 Formal verification and reproducibility

20.1 Public Lean bank

Public repository `main` presently contains the exact problem predicate, the finite exclusion through 4×10^9 , the `WindowPairSupply` closure compiler, the V14 structural source bank, and at least some individual V15 source modules such as `AffineMuLogHardSource.lean`. The aggregate `Main.lean` import spine and V15 status file are not yet synchronized. The claims labelled `PUBLIC LEAN-CHECKED` are therefore limited to declarations directly verified on public `main`; the full V15 replay status below is tracked separately.

20.2 Latest supplied V15 Aristotle replay

A newer user-supplied Aristotle archive and build report contain the following V15 files:

- `AffineMuLogIdentity.lean`;
- `AffineMuLogHardSource.lean`;
- `AffineMuLogLine.lean`;
- `AffineMuLogExponentLedger.lean`;
- `BalancedSevenFinite.lean`;
- `BalancedSevenPolarization.lean`;
- `Erdos287V15Status.lean`.

The report states:

- `lake build` succeeded with 8118 jobs and no errors;
- a Lean-code scan found no `sorry`, `admit`, user `axiom`, `opaque`, `unsafe`, `native_decide`, or `@[implemented_by]`;
- `#print axioms` for the principal new theorems used only subsets of `propext`, `Classical.choice`, and `Quot.sound`;
- the analytic structures `PolarizedOmega7SignedEoD` and `MuLogComparisonLowCondMatch` were deliberately left uninhabited.

Accordingly this candidate labels the exact V15 algebra **LEAN-PROVED IN LATEST V15 REPLAY**. At the time of this editorial audit, public `main` still did not display that full replay, so public synchronization is kept as a separate provenance fact rather than silently conflated with kernel status.

20.3 Factorial Euler layer

The latest supplied V16 factorial Aristotle replay reports the factorial Euler polarization, exact repeated-prime encoding, formal local Euler-series/generalized- Λ identities, expected-term linearity, and the Pascadi parameter ledger as kernel-checked. The build report states 8123 jobs, zero errors, no unfinished-proof or unsafe-code markers, and principal axiom reports restricted to the standard logical principles `propext`, `Classical.choice` and `Quot.sound`. Because the exact factorial replay has not been independently synchronized into every public aggregate status file, this paper labels it **LEAN-PROVED IN LATEST V16 FACTORIAL REPLAY** rather than silently equating replay status with public-main synchronization.

20.4 Prepublication state

This R4 candidate preserves the clean front half, inserts a self-contained Gate 0–2 theory and verification manual, and retains the organized current frontier ledger. It is not a GitHub publication candidate until the current Gate-1B research ledger and any newer Aristotle replay are re-audited.

This file is a local prepublication candidate. It has not been pushed to the canonical GitHub PDF or the legacy forum PDF path. The existing public V16 and the immutable V15 archive remain untouched while this candidate is reviewed.

21 Conclusion

The strongest unconditional result in the project remains finite: there is no exact counterexample to Erdős Problem #287 with maximum denominator at most 4×10^9 . The exact global remaining interface is equally clear: establish effective eventual `WindowPairSupply`.

On the analytic side, the source-minimal $\Lambda = \mu * \log$ algebra and determinant-one line have now been reported kernel-checked in the latest V15 replay. For the balanced-seven cell, the controlling representation is no longer the squarefree polarization plus a collision remainder. The factorial Euler function

$$F_{\mathbf{z}}(p^e) = a_{\mathbf{z}}(p)^e / e!$$

recovers the ordered seven labelled source exactly, including repeated primes, and polarization commutes with every linear expected-term operator.

The first exact V-branch analytic open is therefore

$$\boxed{\text{AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45}},$$

together with the source identification

$$\boxed{\text{AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45}}.$$

The endpoint theorem is much narrower than a general class-C Bombieri–Vinogradov statement. The current DGS and Pascadi theorems do not literally furnish it, and the direct written Pascadi proof parameters are incompatible with $(Q, y) = (X^{3/5}, X^{1/7})$.

Closing those two balanced-seven inputs would still leave the remaining V-cells and M-branch, the literal Ford-generated packet census and provider assignment, the $k = 0$ smooth packet, any balanced- R_9 physical $G(d; n)$ specialization actually generated by the sieve, fixed-certificate leakage/FCL, the source-specific N_2 and Gate-2 lower-bound compiler, effectivity, and eventual large- M supply. Gate 1A is required only if the literal census routes a needed packet to its common-weight provider. Erdős Problem #287 therefore remains open.

Part II

Current Frontier and Provenance Ledger

22 Purpose, coverage baseline and status firewall

Part II restores the *technical depth* of the earlier public dossier without restoring its chronological patch-stack format. Its coverage floor is the V15 public-review snapshot committed as

7fc3797105519c4e6a1669f174ce7394336bdc18,

which published the former 50-page-style Gate-1A/Gate-1B/Ford programme together with the then-current balanced-seven update. That snapshot is used here only as a coverage baseline: when later work changes the controlling route, the later status wins.

Three time/status layers are kept separate throughout:

1. **public/formal bank**: theorem or compiler actually present in the public Lean project;
2. **latest supplied research ledger**: a mathematically audited frontier supplied after the V15 snapshot but not silently promoted to a public Lean or published theorem;
3. **open/source interfaces**: deliberately unproved analytic or source-identification obligations.

This is important for Gate 1B. The public formal socket still records the one-completion coefficient

$$\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p)$$

and the exact 1/104 distribution-margin budget. That socket is neither false nor erased. However, the latest supplied analytic work has moved beyond it to a Motohashi/rank-one frontier. The formal socket is therefore a *banked abstraction/alternative provider interface*, while the rank-one chain below is the current research frontier.

Detailed dead-route archaeology remains outside the main narrative. What is restored here is every current component needed to reconstruct the proof programme: Gate 1A source class, Gate 1B current frontier and downstream firewall, the Ford-generated packet census, balanced- R_9 /fixed-certificate leakage, Gate 2, and the relation of the current factorial- Ω_7 source to those providers.

23 Gate 1A: current canonical-provider ledger

Gate 1A is a conditional theorem for one canonical common-weight physical operator. It is not a universal theorem for arbitrary row-dependent source weights and it is not assumed globally necessary for Erdős #287.

For a row

$$e = (r, k, m), \quad m' = m + kr,$$

the affine forms satisfy

$$m' A_e(t) - m B_e(t) = 2k,$$

and the canonical physical operator is

$$C_e(b, d) = \sum_t W_D(t) \left(\sum_{p \preceq L} b_p \rho_p(A_e(t)) \right) \left(\sum_{q \preceq L} d_q \rho_q(B_e(t)) \right).$$

The conditional target remains

$$\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \sum_{p \preceq L} |b_p|^2 \sum_{q \preceq L} |d_q|^2,$$

with the same normalization conventions as in Part I.

23.1 What is banked and what is still external

Gate-1A component	Status	Current interpretation
Determinant/row algebra and all- m row family	BANKED FINITE/OPERATOR	The theorem is not restricted to clean- P_3 rows.
Canonical common physical weight W_D	SOURCE FIREWALL BANKED	Later row dependence must be deterministically derived from this one datum.
Scale ledger, row-energy conservation and normalization bridge	BANKED	Includes the H^2 conversion and the signed vertex margins.
Finite BPP participation-to-energy compiler	BANKED	Participation implies a moving-family energy saving once the physical participation input is supplied.
Projective ratio-class/axis algebra	BANKED FINITE ALGEBRA	The state diagonal is routed inside the zero-curvature/projective branch rather than discarded.
Outer-square-root algebra and exponent budget	BANKED	The formal compiler checks the square-root conversion but does not manufacture the physical energy pin.
Poisson–Bruhat transport/source ceiling	SOURCE/ANALYTIC PIN	Must preserve the literal physical packet and absolute comparator.
Fixed-state exclusion and prime participation	EXTERNAL/SOURCE PIN	The finite compiler is present; the physical smooth-envelope/short-interval input is external.
Outer-root physical energy identification	SOURCE OPEN	Identifies the abstract BPP energy with the actual Gate energy.
Projective/diagonal physical source routing	SOURCE OPEN	Finite algebra exists; the physical source ceiling must be supplied at target scale.
Literal recombination estimate	SOURCE OPEN	Controls discarded/excision/recombination errors without spending the same saving twice.
Arbitrary edge-dependent D_2 family	OUTSIDE CANONICAL THEOREM	Requires an upstream Gate-0 adapter or finite-template/nuclear reduction.

The current source-glue issue can be summarized as a synchronized two-view/fibre problem: the same physical packet has to satisfy the common-weight representation, the BPP participation representation and the final source-routing representation simultaneously. We refer to this compactly as the *SB- ν /source-glue interface*. It is a source-identification problem, not an exponent deficit in the already-banked finite operator algebra.

23.2 When Gate 1A is required for #287

The correct logical statement is

GATE1A_REQUIRED is decided by the literal Ford-generated packet census.

An actual generated packet may use Gate 1A only after a scale/source dictionary supplies:

- the relation between the Gate ambient scale X_1 and the #287 scale X ;
- the identification of the physical bilinear variables with the Gate row variables;
- the induced R, L, H, K, U, D ranges;
- preservation of the one common physical weight, or a proved finite-template/nuclear reduction with acceptable total cost;
- survival of the binding fixed-power margin after truncation, multiplicity and scale conversion.

No global Gate-1A closure is assumed if the generated census never routes a needed packet to this source class.

24 Gate 1B: from the banked one-completion socket to the current rank-one frontier

Gate 1B is the complementary signed/parity-breaking provider. Its defining feature is that the Möbius sign and shifted congruence data must remain attached to the physical source; squaring or replacing the source by an arbitrary coefficient family can destroy the resource that is meant to break parity.

24.1 Banked one-completion socket

The public challenge file records the authoritative one-completion coefficient

$$\beta_{D,P}(d,p) = \mu_D(d)\Lambda_P(p)$$

as separately linear in the Möbius and prime variables. It also proves the exact dyadic budget

$$\theta_{\text{req}} = \frac{1}{2} + \frac{1}{104}$$

and the implication

literal source-to-convolution-BV interface
at level $\theta \geq \theta_{\text{req}}$
 $\Downarrow$
abstract Gate-1B convolution conclusion

The source-to-interface theorem is deliberately open. This formal socket remains useful, but it no longer describes the first research-level analytic obstruction.

24.2 Latest supplied Motohashi/rank-one chain

The latest Gate-1B research ledger supplied for this candidate gives the following controlling progression. These labels are *research-status labels*, not claims of public Lean or published theorems.

Node	Latest status	Role
MOTOHASHI-ABC-EXACT-PIN45	PASS	Exact A/B/C source pin for the Motohashi decomposition.
TWISTED-DEFECT-ABC45	PASS	Carries the centred/twisted defect data through the A/B/C split.
FIVEFOLD-MOTOHASHI-ITERATION45	PASS	Iterates the exact five-defect source without replacing it by unrelated coefficient norms.
RANKONE-POLYLOGK-INTERIOR45	CLOSED MODULO COMPARISON	Fixed-polylogarithmic- k interior reduced to the comparison/main-term convention.
RANKONE-ENDPOINT-U-DIAGONAL45	POWER CLOSED	Endpoint u -diagonal carries a saving $X^{-1/12+o(1)}$.
RANKONE-ENDPOINT-U-OFFDIAG45	FIRST ANALYTIC OPEN	Current first analytic frontier in the rank-one endpoint family.
RANKONE-HIGHK45	OPEN	High- k regime is not covered by the fixed-polylogarithmic- k interior analysis.
PURE5-COMPARISON-MAINTERM-PIN	SOURCE / COMP. OPEN	Literal physical main-term normalization must match the pure-five source.

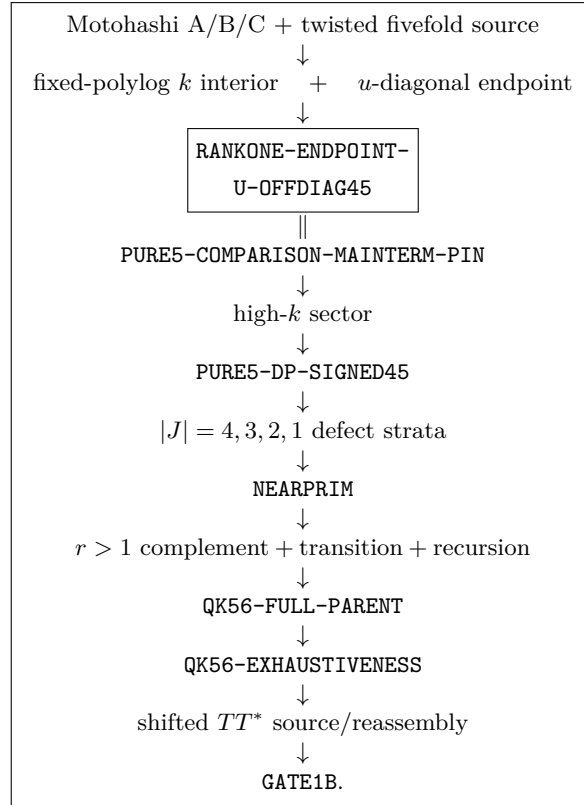
Thus the present first research question is no longer the abstract 1/104 socket and no longer a broad “full covariance” theorem. It is the source-preserving endpoint off-diagonal problem

RANKONE-ENDPOINT-
U-OFFDIAG45

The comparison pin is logically parallel: closing the off-diagonal estimate without the physical main-term convention does not close the pure-five packet.

24.3 Downstream Gate-1B firewall

Even a solution of the first endpoint off-diagonal does not jump directly to Gate 1B. The current source ledger keeps the following downstream structure explicit:



This diagram is a *dependency firewall*: later nodes are not declared solved merely because an earlier parent is controlled.

24.4 Where QK56 sits

The QK56 parent is a full covariance object of the form

$$C_{QK} = \sum_{q_1, q_2} \sum_{h_1, k_1, h_2, k_2} A(q_1, q_2; h_1, k_1, h_2, k_2) Q(q_1; h_1, k_1) Q(q_2; h_2, k_2).$$

The split $q_1 = q_2$ versus $q_1 \neq q_2$ is the same-modulus/cross-modulus split of one parent, not two unrelated theorems. Earlier work established substantial same- q and cross- q children and the canonical zero mode at the finite/algebraic level. The present rank-one chain places QK56 later in the exhaustiveness/reassembly firewall rather than at the first analytic open.

The exact status therefore is:

QK56 remains structurally live, but
**RANKONE-ENDPOINT-
 U-OFFDIAG45**
 is earlier in the current closure order

25 Ford–Maynard generated packets: census before provider

The Ford–Maynard universal Type-II hypothesis is a clean published sufficient wrapper, but Proposition 7.22 does not invoke arbitrary coefficient pairs out of thin air. After a finite decomposition it generates the grouped equation-(7.23) family

$$\sum_{n=n'n'' \asymp x} Y_1(n') Y_2(n'') w_{n'n''}.$$

A proof-specific replacement is therefore legitimate only after two source obligations:

1. a *literal generated-packet census* listing every coefficient family produced by the proof;
2. a replacement certificate C_{FM} proving that every Type-II invocation is covered by one proved provider after all Mellin/Perron, support and norm transformations.

The programme labels

SOURCE-ADAPTER45,
FM-TO-GATE-COORDINATE-CENSUS45

refer to this source dictionary. Neither can be replaced by the slogan “the packet looks Type-II”.

25.1 Current provider-facing census

Generated family	Literal status	Candidate provider	Current blocker
Möbius singleton	source-specific generated packet	Gate 1B / signed source	literal dictionary and stability under generated outer factors
Model singleton	source-specific generated packet	QK56/direct affine completion	native packet dictionary; short Möbius carrier may sit in the complement
Balanced-seven V-cell	explicit fixed-degree source	factorial- Ω_7 endpoint theorem	signed endpoint + physical comparison identification
Balanced R_9 leakage	Ford-generated leakage candidate	fixed-certificate/FCL lane	literal $G(d; n)$ specialization and source reassembly
Common-weight/edge packets	census-dependent	Gate 1A only after exact match	scale/source adapter and common-weight preservation
Pure-five/high-order packets	Gate-1B source family	Motohashi/rank-one chain	endpoint off-diagonal, comparison, high- k , downstream strata
Remaining fixed- g_* packets	not exhaustively censused	TBD from literal packet shape	finite census and provider assignment

This table encodes the main logical correction:

provider assignment follows the generated source; it is not imposed globally.

In particular, Gate 1A is neither deleted nor assumed universally necessary.

26 Balanced R_9 , fixed-certificate leakage and the Gate-2 landing

The balanced order-nine identity

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70$$

is a genuine finite/binomial fact. The formal bank also records the complementary alternating identity. What is *not* automatically proved is that Ford’s actual arithmetic weight $G(d; n)$ selects exactly this divisor-order specialization on the physical balanced- R_9 cell.

Therefore the correct status is:

balanced R_9 = Ford-generated leakage candidate,

with

$H(n) = 70$ only after the literal $G(d; n)$ specialization is proved.

The old R_9 death certificate and the abandoned unbalanced-1/5 redesign are not revived.

The fixed-certificate architecture uses one explicit positive Ford certificate g_* and its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Its transference identity separates:

- total correlation;
- leakage outside the prime/main-composite regions;
- the exceptional N_2 collar;
- the positive comparison margin.

Balanced R_9 is one generated leakage family that can feed this architecture once the physical source dictionary is proved; it is not the whole fixed-certificate leakage problem.

26.1 Gate 2 consumes, rather than invents, the packet bounds

Gate 2 is the Ford–Maynard lower-bound/endgame compiler. It fixes one compatible ε_* and the shrunk parameter triple

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Its source-specific replay requires:

1. Gate-0 comparison regularity, growth and exact Type-I input;
2. the literal $G(d; n)/H(n)$ dictionary and the Ford-generated packet census;
3. the source-specific aggregate N_2 estimate;
4. one compatible fixed ε_* ;
5. either the universal Type-II wrapper, or all generated-(7.23) packet estimates together with **SOURCE-ADAPTER45** and C_{FM} .

Thus the correct closure order is

packet census $\rightarrow$ Gate providers $\rightarrow$ generated Type-II reassembly $\rightarrow$ Gate 2 lower bound,

not Gate 2 followed by a later packet census.

27 Current #287 fixed-degree-seven direction inside the gate programme

Part I gives the controlling factorial Euler encoding

$$F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}, \quad a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p),$$

which exactly includes repeated primes. The first V-branch analytic residual is

AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45

and the physical comparison identification

AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45

remains separate.

The direct equal-seven three-block dictionary into Pascadi Proposition 6.3 fails its range geometry, and the direct smooth-number/multiplicative-function proof architecture is nonclosing at $y = X^{1/7}$ for the exact parameter reason proved in Part I. The current research direction is therefore *post-dispersion fixed-degree seven*: keep the degree seven source literal, perform the relevant dispersion/completion first, and search for a four-block or comparable structured theorem only after the physical source has exposed the correct variables. This is a research direction, not a theorem or an already-identified published provider.

If the factorial endpoint and comparison pin are both proved, the balanced-seven cell closes. That implication is deliberately local:

Balanced7 closed $\not\Rightarrow$ FCL closed $\not\Rightarrow$ WindowPairSupply.

The remaining packet census and Gate-provider assignment still intervene.

28 Master current-status ledger

Node	Current status	Publication interpretation
Finite exclusion $M \leq 4 \times 10^9$ WindowPairSupply compiler	PUBLIC LEAN-CHECKED PUBLIC LEAN-CHECKED / CONDITIONAL	strongest unconditional theorem exact global large- M interface
Gate 0 comparison/growth framework	MIXED PASS / SOURCE PINS	literal Type-I/publication dictionary still matters
Gate 1A finite/operator mechanism	SUBSTANTIALLY BANKED	conditional canonical common-weight provider
Gate 1A physical five-pin/source glue	OPEN SOURCE/ANALYTIC	needed only for packets censused into Gate 1A

Node	Current status	Publication interpretation
Gate 1B one-completion 1/104 socket	BANKED FORMAL INTERFACE	valid abstraction, not first research frontier
Motohashi A/B/C and twisted fivefold source	LATEST RESEARCH PASS	current Gate-1B source reduction
Rank-one polylog- k interior	CLOSED MODULO COMPARISON	latest supplied research ledger
Rank-one u -diagonal endpoint	POWER CLOSED $X^{-1/12+o(1)}$	latest supplied research ledger
RANKONE-ENDPOINT-U-OFFDIAG45	FIRST GATE-1B ANALYTIC OPEN	current rank-one frontier
RANKONE-HIGHK45	OPEN	separate high- k regime
PURE5-COMPARISON-MAINTERM-PIN	SOURCE OPEN	parallel normalization requirement
Pure-five $\rightarrow$ defect strata $\rightarrow$ near-primitive chain	DOWNSTREAM OPEN/ROUTING	must not be skipped after rank-one closure
QK56 full parent/exhaustiveness	LATER GATE-1B FIREWALL	structurally live, not first current open
Ford generated-(7.23) census	OPEN SOURCE/FINITE CENSUS	decides provider assignment
SOURCE-ADAPTER45, C_{FM}	OPEN	required for proof-specific replacement of universal Type II
Balanced R_9 binomial 70	FINITE PASS	physical $G(d; n)$ specialization still source open
Factorial Euler Ω_7 encoding	RESEARCH PASS / LEAN PENDING	current balanced-seven representation
Factorial signed endpoint	OPEN ANALYTIC	current exact V-branch residual
Factorial/physical comparison match	OPEN SOURCE	principal local density + literal exceptional-character convention
Remaining fixed- g_* census	OPEN	balanced-seven is only one generated family
FCL / positive affine mass	OPEN	requires all leakage/provider inputs
Gate 2 lower-bound replay	CONDITIONAL / NOT ACTIVATED	consumes generated packet bounds, $N_2, G/H, \varepsilon_*$
Eventual large- M supply/effectivity	OPEN	needed before the finite Lean compiler applies globally
Erdős Problem #287	OPEN	no unconditional proof claim

29 Superseded, false and provenance-only routes

The following are preserved for auditability but are not allowed to displace the current frontier:

- the V15 squarefree polarization plus $X^{6/7+o(1)}$ repeated-prime router: valid historical representation, superseded as controlling by the factorial Euler encoding;
- complete multiplicativity as a required repair: *false*; ordinary multiplicativity in class $\mathcal{C}$ is enough for the relevant dictionary;
- a stated Pascadi “Theorem 1.7” multiplicative-function black box: not present in the audited current arXiv v2;
- the direct Pascadi v2 proof route at $(Q, y) = (X^{3/5}, X^{1/7})$: audited nonclosing because $\eta \leq 1/4000$ conflicts with $\eta \geq 1/7$;
- the V14 Vaughan/prime-modulus Möbius two-outer route: mathematically valid but nonminimal, retained as source audit/provenance;
- direct singleton = existing Gate-1B common-conductor covariance: *false* as a literal source dictionary;
- direct canonical Gate-1A determinant adapter for the affine determinant-one source: fails literally ($2k$ versus ± 1);
- equal-seven Pascadi Proposition 6.3 three-block dictionary: *false* at the literal equal-seven geometry;
- ordinary separated multiplicative large sieve / separated high moments: nonclosing by the previously audited $X^{1/10}$ deficit;
- modulus micro-switch as an exhaustive closure: nonexhaustive because the prime-modulus sector survives.

The public paper therefore retains V15’s *diagnostic density* while keeping obsolete routes visually and logically subordinate to the current closure chain.

A Formal theorem and provenance map

Public Lean bank

- `RequestProject/Erdos287/ProblemStatement.lean`: exact problem predicate and ordered public-statement compiler.
- Finite-range files: adjacent-hole obstruction, prime-power window blocker, and `no_Erdos287Counterexample_of_max_le_4e9`.
- `RequestProject/Erdos287/ClosureInputs.lean`: `WindowPairSupply` and the end-to-end conditional closure compiler.
- V14 source files/status: valid Vaughan structural audit, retained as provenance rather than the controlling source-minimal route.

Lean-proved in the latest supplied V15 replay

The supplied V15 build report records kernel-checked proofs of:

- $\Lambda = \mu * \log$ and its affine specialization;
- exhaustive small- q /small- r /hard recombination;
- the determinant-one line, coprimality, iff parametrization, and uniqueness;
- the finite exponent ledger;
- the balanced-seven binomial coefficient -20 ;
- the repeated-prime finite router;
- the squarefree labelled polarization and its 7^{-7} normalization.

The same status source leaves the analytic endpoint and comparison structures uninhabited.

Research-proved / Lean pending

The current hostile audit adds:

- factorial Euler polarization $F_{\mathbf{z}}(p^e) = a_{\mathbf{z}}(p)^e/e!$;
- exact ordered coefficient extraction including repeated primes;
- $F_{\mathbf{z}} \in \mathcal{C}$ from its Euler logarithmic derivative;
- exact commutation with every linear expected-term operator;
- the proof-parameter obstruction $\eta \leq 1/4000$ versus $\eta \geq 1/7$ for the direct Pascadi v2 route.

The latest supplied V16 factorial replay is claimed as kernel-checked; public-main synchronization remains separate.

Gate and Ford–Maynard source provenance

The Gate architecture in Section 4 is a compressed restoration of the earlier standalone verification dossier, not a reconstruction from theorem names alone. The latest supplied formal bank further records: the Gate-1A abstract synchronized two-view fibre challenge $N_{\nu} = X^{o(1)}$ together with unit-twist portability; the Gate-1B one-completion coefficient $\beta_{D,P} = \mu_D \Lambda_P$, fixed-unit reciprocal-source invariance, the exact $1/104$ dyadic margin, and the open source-to-convolution-BV interface; and the balanced- R_9 finite binomial certificate. The physical Ford $G(d;n)$ specialization, the generated- (7.23) census, the replacement certificate C_{FM} , the Gate-1A necessity decision and all Gate-2 analytic inputs remain open.

Historical technical provenance

The following remain useful diagnostics but are not part of the controlling public proof architecture:

- the squarefree polarization plus the $X^{6/7+o(1)}$ repeated-prime router;
- the V14 Vaughan/prime-modulus Möbius two-outer route;
- failed direct Gate-1A and native-QK dictionaries;
- separate multiplicative-large-sieve and equal-seven three-block death tests.

They belong in the technical verification dossier rather than in the main narrative.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique, 1980, p. 33.
- [2] T. F. Bloom, *Erdős Problem #287*, Erdős Problems archive, problem provenance and status page, accessed August 2026.
- [3] K. Ford and J. Maynard, *On the theory of prime producing sieves*, arXiv:2407.14368, 2024.
- [4] S. Drappeau, A. Granville and X. Shao, *Smooth-supported multiplicative functions in arithmetic progressions beyond the $x^{1/2}$ -barrier*, *Mathematika* 63 (2017), 895–918.
- [5] A. Pascadi, *On the exponents of distribution of primes and smooth numbers*, arXiv:2505.00653v2, 29 June 2025.
- [6] The Lean and Mathlib communities, *Lean 4 and Mathlib*, formal proof assistant and mathematical library.